

Text Data in Business and Economics

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5. Unsupervised and Supervised Learning from Text

Different Goals, Different Methods

- ▶ Unsupervised Learning
 - ▶ algorithm discovers themes/patterns in text (or other high-dimensional data)
 - ▶ human interprets the results (e.g. inspect content of topics or clusters)

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 - ▶ machine learns to replicate labels for new data points
- ▶ Both strategies amplify human effort, each in different ways
- ▶ Distinctions are not clear-cut:
 - ▶ unsupervised learning models can be used in service of prediction or known goals
 - ▶ supervised learning models can be used to discover themes/patterns

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4. Empirical analysis
 - ▶ Produce statistics or predictions with the trained model
 - ▶ **Answer the research question**

Outline

Dimensionality Reduction

Topic Models

Supervised Learning

- Overview

- Regression / Regularization

- Binary Classification

- Multi-Class Models

The Document-Term Matrix is high-dimensional

The **document-term matrix** $\mathbf{X}$:

- ▶ each row d represents a **document**, while each column w represents a word (or term more generally, e.g. n-grams).
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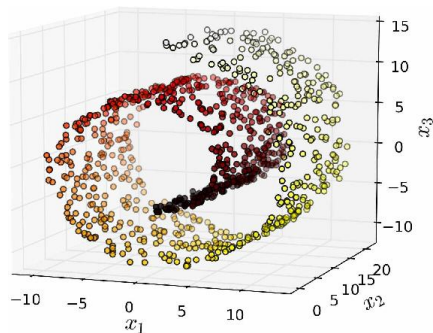
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→ $\mathbf{X}$ often has billions of cells.

Distribution of Datasets

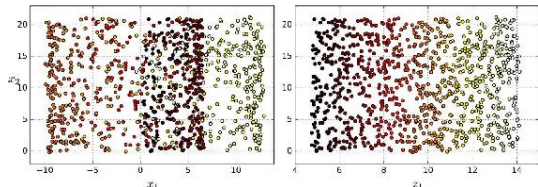
- ▶ Datasets are not distributed uniformly across the feature space
- ▶ They have a lower-dimensional latent structure – a **manifold** – that can be learned

“The Swiss Roll”

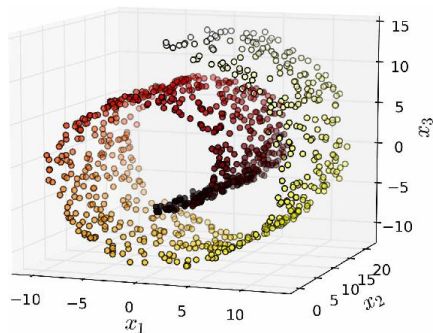


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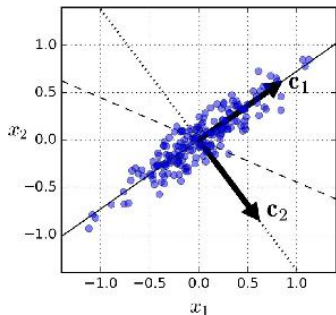
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- ▶ **Dimensionality reduction** makes data more interpretable – for example by projecting down to two dimensions for visualization
- ▶ improves computational tractability
- ▶ can improve model performance

What dimension reductions have you already tried in this class?

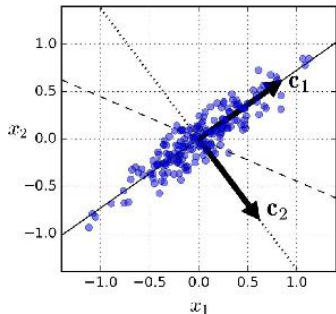
PCA (principal component analysis)



- PCA computes the dimension in data explaining most variance.

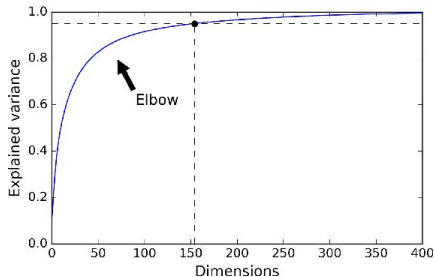
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- after the first component, subsequent components learn the (orthogonal) dimensions explaining most variance in dataset after projecting out first component

PCA and LSA

The document-term matrix $\mathbf{X}$ can be reduced by projecting down to first principal component dimensions

- ▶ This is known as “latent semantic analysis”
- ▶ Distance metrics between observations (e.g. cosine similarity) are approximately preserved

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The document-term matrix $\mathbf{X}$ can be reduced by projecting down to first principal component dimensions

- ▶ This is known as “latent semantic analysis”
- ▶ Distance metrics between observations (e.g. cosine similarity) are approximately preserved
- ▶ PCA factors are not interpretable
 - ▶ For non-negative data (e.g. counts or frequencies), **Non-negative Matrix Factorization (NMF)** provides more interpretable factors than PCA

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 - ▶ tell a story not just about what, but how and why
 - ▶ **topic models are more interpretable** than other dimension reduction methods, such as PCA

Latent Dirichlet Allocation

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Latent Dirichlet Allocation

- ▶ Each topic is a distribution over words
- ▶ Each document is a distribution over topics
- ▶ Input: $N \times M$ document-term count matrix X
- ▶ Like NMF, LDA works by factorizing X into:
 - ▶ an $N \times K$ document-topic matrix
 - ▶ an $K \times M$ topic-term matrix
- ▶ Assume: there are K topics (tunable hyperparameter, use coherence)
- ▶ Unlike NMF, LDA is probabilistic:
 - ▶ Goal: Estimate the latent (hidden) topic structure that best explains the word co-occurrences in documents
 - ▶ Algorithm maximizes the likelihood of observed documents given the latent topic structure
 - ▶ Output: Estimates of probabilities that each word and each document are composed of each topic
 - ▶ LDA is a generative probabilistic model, meaning it assumes documents were generated by an underlying topic structure, and it tries to reverse-engineer that structure
 - ▶ Dirichlet priors: Hyperparameters for topics per document (alpha), and words per topic (beta)

Using an LDA Model

Once trained, can easily get topic proportions for a corpus

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Can then use the topic proportions as variables in a economics analysis.

- ▶ e.g., Catalinac (2016) shows that after a Japanese political reform that reduced intraparty competition, candidate platforms reduced pork-barrel policies and increased national ones

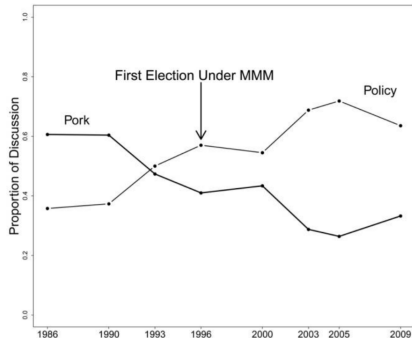


TABLE 1 A Summary of Common Assumptions and Relative Costs Across Different Methods of Discrete Text Categorization

	Method				
	<i>Reading</i>	<i>Human Coding</i>	<i>Dictionaries</i>	<i>Supervised Learning</i>	<i>Topic Model</i>
A. Assumptions					
<i>Categories are known</i>	No	Yes	Yes	Yes	No
<i>Category nesting, if any, is known</i>	No	Yes	Yes	Yes	No
<i>Relevant text features are known</i>	No	No	Yes	Yes	Yes
<i>Mapping is known</i>	No	No	Yes	No	No
<i>Coding can be automated</i>	No	No	Yes	Yes	Yes
B. Costs					
Preanalysis Costs					
<i>Person-hours spent conceptualizing</i>	Low	High	High	High	Low
<i>Level of substantive knowledge</i>	Moderate/High	High	High	High	Low
Analysis Costs					
<i>Person hours spent per text</i>	High	High	Low	Low	Low
<i>Level of substantive knowledge</i>	Moderate/High	Moderate	Low	Low	Low
Postanalysis Costs					
<i>Person-hours spent interpreting</i>	High	Low	Low	Low	Moderate
<i>Level of substantive knowledge</i>	High	High	High	High	High

Recommended: read this part of Quinn, Monroe, Colaresi, Crespin, and Radev (2010).

Structural Topic Model = LDA + Metadata

Roberts, Stewart, and Tingley

STM provides two ways to include contextual information:

- ▶ Topic prevalence can vary by metadata
 - ▶ e.g. Republicans talk about military issues more than Democrats

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- ▶ Topic prevalence can vary by metadata
 - ▶ e.g. Republicans talk about military issues more than Democrats
- ▶ Topic content can vary by metadata
 - ▶ e.g. Republicans talk about military issues more patriotically than Democrats
- ▶ Structural topic model is not a prediction model:
 - ▶ it will tell you which topics or features correlate with an outcome, but it will not provide an in-sample or out-of-sample prediction for an outcome
- ▶ It actually uses another distribution of the priors (*not* Dirichlet) such that without covariates it replicates the correlated topic model (Blei and Lafferty, 2005)

Recent Advances in Topic Model

- ▶ Keyword-Assisted topic model (Eshima, Imai, and Sasaki, 2024)
 - ▶ allows *semi-supervised* creation of topics
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 - ▶ allows *semi-supervised* creation of topics
 - ▶ input seed dictionaries and labeled topics
- ▶ Problems with unstructured data and causal inference (Battaglia et al., 2024)
 - ▶ shows that two-steps strategy leads to invalid inference
 - ▶ propose solutions with bias correction and a one-step strategy

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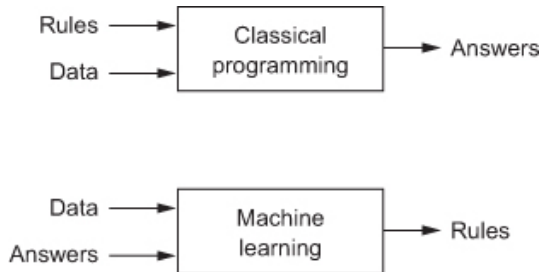
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What is supervised machine learning?



- ▶ In classical computer programming, humans input the rules and the data, and the computer provides answers.
- ▶ In machine learning, humans input the data and the answers, and the computer learns the rules.

What do ML Algorithms do? Fit a function to data points

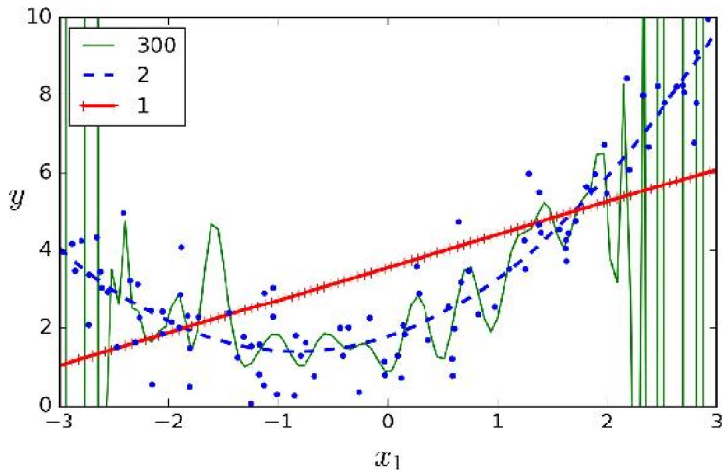


Figure 4-14. High-degree Polynomial Regression

What do ML Algorithms do? Minimize a cost function

- ▶ A typical cost function (or loss function) for regression problems is Mean Squared Error (MSE):

$$\text{MSE}(\theta) = \frac{1}{n_D} \sum_{i=1}^{n_D} (h(x_i; \theta) - y_i)^2$$

- ▶ n_D , the number of rows/observations
- ▶ x , the matrix of predictors, with row x_i
- ▶ y , the vector of outcomes, with item y_i
- ▶ $h(x_i; \theta) = \hat{y}$ the model prediction (hypothesis)

The **data** (x, y) are taken as given, and the ML algorithm searches for **parameters** θ to minimize the cost function

Loss functions, more generally

- ▶ The loss function $L(\hat{\mathbf{y}}, \mathbf{y})$ assigns a score based on prediction and truth:
 - ▶ Should be bounded from below, with the minimum attained only for cases where the prediction is correct
- ▶ The average loss for the test set is

$$\mathcal{L}(\theta) = \frac{1}{n_D} \sum_{i=1}^{n_D} L(h(\mathbf{x}_i; \theta), \mathbf{y}_i)$$

- ▶ The estimated parameter matrix θ solves

$$\hat{\theta} = \arg \min_{\theta} \mathcal{L}(\theta)$$

Linear Regression is Machine Learning

- ▶ Ordinary Least Squares Regression (OLS) assumes the functional form $f(x; \theta) = x'_i \theta$ and minimizes the mean squared error (MSE)

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- ▶ This minimand has a closed form solution

$$\hat{\theta} = (\mathbf{x}'\mathbf{x})^{-1}\mathbf{x}'\mathbf{y}$$

- ▶ most machine learning models do **not** have a closed form solution → use numerical optimization instead (gradient descent).

Gradient Descent

$$\text{MSE}(\theta) = \frac{1}{n_D} \sum_{i=1}^{n_D} (h(\theta; \mathbf{x}_i) - y_i)^2$$

- ▶ The partial derivative for feature j is

$$\frac{\partial \text{MSE}}{\partial \theta_j} = \frac{2}{n_D} \sum_{i=1}^{n_D} \underbrace{(h(\theta; \mathbf{x}_i) - y_i)}_{\text{error for this obs}} \underbrace{\frac{\partial h(\theta; \mathbf{x}_i)}{\partial \theta_j}}_{\text{how } \theta_j \text{ shifts } h(\cdot)}$$

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- ▶ The **gradient** ∇ gives the vector of these partial derivatives for all features:
- ▶ **Gradient descent** nudges θ against the gradient (the direction that reduces MSE):

$$\nabla_{\theta} \text{MSE} = \begin{bmatrix} \frac{\partial \text{MSE}}{\partial \theta_1} \\ \frac{\partial \text{MSE}}{\partial \theta_2} \\ \vdots \\ \frac{\partial \text{MSE}}{\partial \theta_{n_x}} \end{bmatrix}$$

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} \text{MSE}$$

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- ▶ If the cost function is convex, gradient descent is guaranteed to find the global minimum

Gradient Descent

- ▶ even when cost function is not convex (eg neural nets), gradient descent often gets decent results
- ▶ **Stochastic gradient descent (SGD)** computes the gradient for a single randomly sampled data point (at each iteration)
 - ▶ Much faster, still works well

Evaluation: Use Cross-Validation During Model Training

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- ▶ Train/Test Split:
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 - ▶ standard approach: randomly sample 80% training dataset to learn parameters, form predictions in 20% testing dataset for evaluating performance
- ▶ Within the training set:
 - ▶ Use cross-validation with grid search to get model performance metrics across subsets of data using different hyperparameter specs.
 - ▶ Find the best hyperparameters for out-of-fold prediction in the training set
- ▶ Then evaluate model performance in the test set using these hyperparameters

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- ▶ We have a corpus (or dataset) D of $n_D \geq 1$ documents d_i (or data points).
- ▶ Each document i has an associated outcome or label $\mathbf{y}_i$ with dimensions $n_y \geq 1$
- ▶ Some documents are labeled and some are unlabeled $\rightarrow$
 - ▶ we would like to learn a function $\hat{\mathbf{y}}(d_i)$ based on the labeled data ...
 - ▶ ... to machine-classify the unlabeled data.

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- ▶ The solution: all the methods from previous lectures for extracting informative numerical information from documents:
 - ▶ style features
 - ▶ counts over dictionary patterns
 - ▶ tokens
 - ▶ n-grams
 - ▶ principal components
 - ▶ topic shares
 - ▶ etc.
- ▶ documents can thus be **featurized** – represented as a matrix of vectors $\mathbf{x}$ with $n_x \geq 1$ features.

Three Types of (Standard) Machine Learning Problems

Determined by the data type of the outcome variable (or label):

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- ▶ **Binary classification:** two choices, normalized to zero and one.
 - ▶ e.g., guilty or innocent
- ▶ **Multinomial Classification:** Three or more discrete, un-ordered outcomes.
 - ▶ e.g., predict what judge is assigned to a case: Alito, Breyer, or Cardozo

Model Evaluation in Test Set

Evaluating a “good” model is context-dependent. Here are some basics

Regression:

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- ▶ mean absolute error (MAE, $\sum |\hat{y}(\theta) - y|$) is less sensitive to outliers
- ▶ R-squared

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accuracy = (# correct test-set predictions) / (# of test-set observations)
- ▶ What if one of the outcomes is over-represented – e.g., 19 out of 20? Then I can guess the modal class and get 95% accuracy
 - ▶ Some alternative classifier metrics designed to address class imbalance (more below)

Regression models $\leftrightarrow$ Continuous outcome

- ▶ If the outcome is continuous (e.g., Y = tax revenues collected, or criminal sentence imposed in months of prison):
 - ▶ Need a regression model
- ▶ Problems with OLS:
 - ▶ tends to over-fit training data
 - ▶ cannot handle multicollinearity

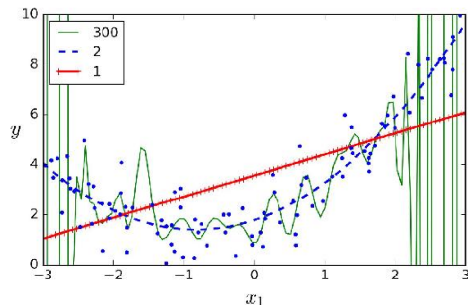


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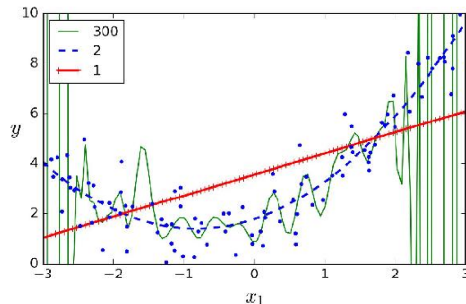


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- ▶ **Regularization:** model training methods designed to reduce/prevent over-fitting

Regularized Loss Function

$$\hat{\theta} = \arg \min_{\theta} \frac{1}{n_D} \sum_{i=1}^{n_D} L(h(\mathbf{x}_i; \theta), \mathbf{y}_i) + \lambda R(\theta)$$

- ▶ $R(\theta)$ is a “regularization function” or “regularizer”, designed to reduce over-fitting
- ▶ λ is a hyperparameter where higher values increase regularization

Regularized Loss Function

$$\hat{\theta} = \arg \min_{\theta} \frac{1}{n_D} \sum_{i=1}^{n_D} L(h(\mathbf{x}_i; \theta), \mathbf{y}_i) + \lambda R(\theta)$$

- ▶ $R(\theta)$ is a “regularization function” or “regularizer”, designed to reduce over-fitting
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In particular:

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Binary Outcome $\leftrightarrow$ Binary Classification

- ▶ Binary classifiers try to match a boolean outcome $y \in \{0, 1\}$.
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 - ▶ Prediction rule is 0 for $\hat{y} < .5$ and 1 otherwise.

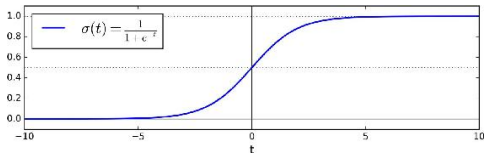
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- ▶ The binary cross-entropy (or log loss) is:

$$L(\theta) = \underbrace{-\frac{1}{n_D}}_{\text{negative}} \sum_{i=1}^{n_D} \left[\underbrace{y_i}_{y_i=1} \underbrace{\log(\hat{y}_i)}_{\log \text{ prob}_{y_i=1}} + \underbrace{(1-y_i)}_{y_i=0} \underbrace{\log(1-\hat{y}_i)}_{\log \text{ prob}_{y_i=0}} \right]$$

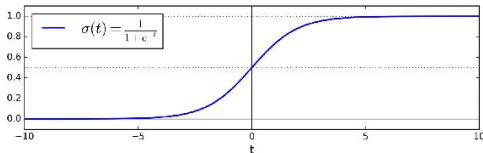
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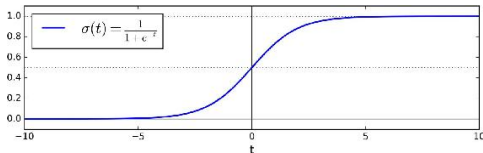
- Plugging into the binary-cross entropy loss gives the logistic regression cost objective:

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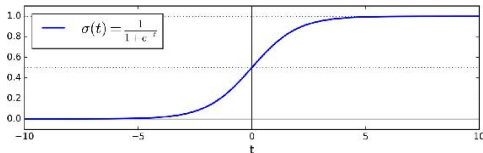
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- Like linear regression, logistic regression can be regularized with L1 or L2 penalties.

Evaluating Binary Classification Models

A **Confusion Matrix** is a nice way to visualize classifier performance:

		Predicted Class	
		Negative	Positive
2*True Class	Negative	# True Negatives	# False Positives
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$$\text{Recall (for positive class)} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

- ▶ Recall decreases with false negatives. “When this outcome occurs, I don’t miss it.”

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AUC-ROC = Area Under the Receiver Operating Characteristic Curve

- ▶ Provides an aggregate measure of performance across all possible classification thresholds
- ▶ False Positive Rate (FPR) vs. True Positive Rate (TPR)
- ▶ Interpretation: randomly sample one positive and one negative example AUC = probability that the model correctly guesses which is which

Multiple Classes: Setup

- The outcome is $y_i \in \{1, \dots, k, \dots, n_y\}$ output classes, which can also be represented as a one-hot vector

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- ▶ We want to learn a vector function

$$\mathbf{y} = \mathbf{h}(\mathbf{x}, \theta)$$

taking text features $\mathbf{x}$ as inputs and outputting a vector of probabilities across outcome classes:

$$\hat{\mathbf{y}} = \{\hat{y}^1, \dots, \hat{y}^{n_y}\}, \sum_{k=1}^{n_y} \hat{y}^k = 1, \hat{y}^k \geq 0 \quad \forall k$$

- ▶ for prediction step, can select the highest-probability class:

$$\tilde{y} = \arg \max_k \hat{y}_{[k]}$$

- ▶ The standard loss function in multinomial classification is **categorical cross entropy**:

$$L(\theta) = - \sum_{k=1}^{n_y} \mathbf{y}^k \log(\hat{y}^k(\mathbf{x}, \theta))$$

Multinomial Logistic Regression

Multinomial logistic regression computes probabilities for each class k using the softmax transformation

$$\hat{y}_k(\mathbf{x}_i) = \Pr(y_i = k) = \frac{\exp(\theta'_k \mathbf{x}_i)}{\sum_{l=1}^{n_y} \exp(\theta'_l \mathbf{x}_i)}$$

- ▶ softmax is the multiclass generalization of sigmoid $\rightarrow$ can then interpret $\hat{y}$ as probabilities.
- ▶ n_x features and n_y output classes $\rightarrow$ there is a $n_y \times n_x$ parameter matrix Θ , where the parameters for each class θ_k are stored as rows.

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The **L2-penalized logistic regression** has loss function

$$\mathcal{L}(\theta) = -\frac{1}{n_D} \sum_{i=1}^{n_D} \log \frac{\exp(\theta'_k \mathbf{x}_i)}{\sum_{l=1}^{n_y} \exp(\theta'_l \mathbf{x}_i)} + \lambda \sum_{j=1}^{n_x} \sum_{k=1}^{n_y} (\theta_{[j,k]})^2$$

- ▶ λ = strength of L2 penalty (could also add lasso penalty)
 - ▶ as before, predictors should be scaled to the same variance.

		Predicted Class		
		Class A	Class B	Class C
3*True Class	Class A	Correct A	A, classed as B	A, classed as C
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More generally, with **multi-class confusion matrix** M with items M_{ij} (row i , column j):

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Can average these metrics across classes to get aggregate metrics

- ▶ e.g., balanced accuracy = unweighted average of recalls across classes
- ▶ can weight classes by their frequency in dataset

Ensemble Methods

Key Idea: Combine multiple models to improve accuracy and reduce errors

- ▶ Voting classifiers (ensembles of different models that vote on the prediction) generally out-perform the best classifier in the ensemble
- ▶ More diverse algorithms will make different types of errors, and improve your ensemble's robustness

Types of Ensemble Methods:

- ▶ Bagging – Combine independent models (Random Forest)
- ▶ Boosting – Sequentially improve weak models (Gradient Boosting: XGBoost)

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6. Answer the research question!